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On general relativity:

python computations.

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Abstract

In these notes, we will compute general relativity with the help of python. We will take the exterior Schwarschild metric, the interior Schwarschild metric and the Friedmann-Robertson-Walker metric. We will calculate the Christoffel symbols, the Riemann tensor, the Ricci tensor and the Einstein tensor.

Chapter 1

Exterior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -\frac{1}{A(r)} & 0 & 0 & 0 \\ 0 & \frac{1}{B(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

1.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} , \quad \Gamma^0_{10} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} , \quad \Gamma^1_{00} = \frac{0.5 \frac{d}{dr} A(r)}{B(r)} ,$$

$$\Gamma^1_{11} = \frac{0.5 \frac{d}{dr} B(r)}{B(r)} , \quad \Gamma^1_{22} = -\frac{1.0 r}{B(r)} , \quad \Gamma^1_{33} = -\frac{1.0 r \sin^2(\theta)}{B(r)} ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} , \quad \Gamma^2_{21} = \frac{1.0}{r} , \quad \Gamma^2_{33} = -0.5 \sin(2\theta) ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} , \quad \Gamma^3_{23} = \frac{1.0}{\tan(\theta)} , \quad \Gamma^3_{31} = \frac{1.0}{r} ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} , \quad .$$

1.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} ,$$

$$R^0{}_{110} = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} ,$$

$$R^0{}_{202} = -\frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} ,$$

$$R^0{}_{220} = \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} ,$$

$$R^0{}_{303} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} ,$$

$$R^0{}_{330} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} ,$$

$$R^1{}_{001} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} ,$$

$$R^1{}_{010} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} ,$$

$$R^1{}_{212} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} ,$$

$$R^1{}_{221} = -\frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} ,$$

$$R^1{}_{313} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} ,$$

$$R^1{}_{331} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} ,$$

$$R^2{}_{002} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} ,$$

$$R^2{}_{020} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} ,$$

$$R^2{}_{112} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} ,$$

$$R^2{}_{121} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^2{}_{323} = \frac{1.0\,(B(r)-1)\sin^2(\theta)}{B(r)} \ ,$$

$$R^2{}_{332} = \frac{1.0\cdot(1-B(r))\sin^2(\theta)}{B(r)} \ ,$$

$$R^3{}_{003} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^3{}_{030} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^3{}_{113} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^3{}_{131} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^3{}_{223} = -1.0 + \frac{1.0}{B(r)} \ ,$$

$$R^3{}_{232} = 1.0 - \frac{1.0}{B(r)} \ ,$$

.

1.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} + \frac{1.0\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R_{11} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R_{22} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 1.0 - \frac{1.0}{B(r)} \ ,$$

$$R_{33} = \frac{\left(0.5rA(r)\frac{d}{dr}B(r) - 0.5rB(r)\frac{d}{dr}A(r) + 1.0A(r)B^2(r) - 1.0A(r)B(r)\right)\sin^2(\theta)}{A(r)B^2(r)} \ ,$$

.

1.4 $(1,1)$ Ricci tensor

The non-vanishing components of the $(1,1)$ Ricci tensor are

$$R^0{}_0 = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{1.0\frac{d}{dr}A(r)}{rA(r)B(r)} ,$$

$$R^1{}_1 = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB^2(r)} ,$$

$$R^2{}_2 = \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} - \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{1.0}{r^2} - \frac{1.0}{r^2B(r)} ,$$

$$R^3{}_3 = \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} - \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{1.0}{r^2} - \frac{1.0}{r^2B(r)} ,$$

.

1.5 Ricci scalar

The Ricci scalar is

$$R = -\frac{1.0\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{2.0\frac{d}{dr}B(r)}{rB^2(r)} - \frac{2.0\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{2.0}{r^2} - \frac{2.0}{r^2B(r)} .$$

1.6 $(0,2)$ Einstein tensor

The non-vanishing components of the $(0,2)$ Einstein tensor are

$$G_{00} = \frac{1.0\left(r\frac{d}{dr}B(r) + B^2(r) - B(r)\right)A(r)}{r^2B^2(r)} ,$$

$$G_{11} = \frac{1.0\frac{d}{dr}A(r)}{rA(r)} - \frac{1.0B(r)}{r^2} + \frac{1.0}{r^2} ,$$

$$G_{22} = \frac{1.0r\left(0.5rA(r)B(r)\frac{d^2}{dr^2}A(r) - 0.25rA(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) - 0.25rB(r)\left(\frac{d}{dr}A(r)\right)^2 - 0.5A^2(r)\frac{d}{dr}B(r\right.}{A^2(r)B^2(r)}$$

$$G_{33} = \frac{1.0r\left(0.5rA(r)B(r)\frac{d^2}{dr^2}A(r) - 0.25rA(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) - 0.25rB(r)\left(\frac{d}{dr}A(r)\right)^2 - 0.5A^2(r)\frac{d}{dr}B(r\right.}{A^2(r)B^2(r)}$$

.

1.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^{0}{}_{0} = -\frac{1.0\frac{d}{dr}B(r)}{rB^2(r)} - \frac{1.0}{r^2} + \frac{1.0}{r^2B(r)} \ ,$$

$$G^{1}{}_{1} = \frac{1.0\left(r\frac{d}{dr}A(r) + (1 - B(r))\,A(r)\right)}{r^2A(r)B(r)} \ ,$$

$$G^{2}{}_{2} = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} + \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} \ ,$$

$$G^{3}{}_{3} = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} + \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} \ ,$$

.

Chapter 2

Interior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -e^{\nu(r)} & 0 & 0 & 0 \\ 0 & e^{\lambda(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -e^{-\nu(r)} & 0 & 0 & 0 \\ 0 & e^{-\lambda(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

2.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = 0.5 \frac{d}{dr}\nu(r) \ , \quad \Gamma^0_{10} = 0.5 \frac{d}{dr}\nu(r) \ , \quad \Gamma^1_{00} = 0.5 e^{-\lambda(r)+\nu(r)} \frac{d}{dr}\nu(r) \ ,$$

$$\Gamma^1_{11} = 0.5 \frac{d}{dr}\lambda(r) \ , \quad \Gamma^1_{22} = -1.0 r e^{-\lambda(r)} \ , \quad \Gamma^1_{33} = -1.0 r e^{-\lambda(r)} \sin^2(\theta) \ ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ , \quad \Gamma^2_{21} = \frac{1.0}{r} \ , \quad \Gamma^2_{33} = -0.5 \sin(2\theta) \ ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} \ , \quad \Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ , \quad \Gamma^3_{31} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ , \quad .$$

2.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$

$$R^0{}_{110} = -0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$

$$R^0{}_{202} = -0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{220} = 0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{303} = -0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{330} = 0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\nu(r) \ ,$$

$$R^1{}_{001} = \left(0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r)\right)e^{-\lambda(r)+\nu(r)} \ ,$$

$$R^1{}_{010} = \left(-0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r)\right)e^{-\lambda(r)+\nu(r)} \ ,$$

$$R^1{}_{212} = 0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$

$$R^1{}_{221} = -0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$

$$R^1{}_{313} = 0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\lambda(r) \ ,$$

$$R^1{}_{331} = -0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\lambda(r) \ ,$$

$$R^2{}_{002} = -\frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$

$$R^2{}_{020} = \frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$

$$R^2{}_{112} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$

$$R^2{}_{121} = \frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$

$$R^2{}_{323} = 1.0\left(e^{\lambda(r)} - 1\right)e^{-\lambda(r)}\sin^2(\theta)\ ,$$

$$R^2{}_{332} = 1.0\cdot\left(1 - e^{\lambda(r)}\right)e^{-\lambda(r)}\sin^2(\theta)\ ,$$

$$R^3{}_{003} = -\frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r}\ ,$$

$$R^3{}_{030} = \frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r}\ ,$$

$$R^3{}_{113} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r}\ ,$$

$$R^3{}_{131} = \frac{0.5\frac{d}{dr}\lambda(r)}{r}\ ,$$

$$R^3{}_{223} = -1.0 + 1.0e^{-\lambda(r)}\ ,$$

$$R^3{}_{232} = 1.0 - 1.0e^{-\lambda(r)}\ ,$$

.

2.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) + 1.0\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)+\nu(r)}}{r}\ ,$$

$$R_{11} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) + \frac{1.0\frac{d}{dr}\lambda(r)}{r}\ ,$$

$$R_{22} = \left(0.5r\left(\frac{d}{dr}\lambda(r) - \frac{d}{dr}\nu(r)\right) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}\ ,$$

$$R_{33} = \left(0.5r\frac{d}{dr}\lambda(r) - 0.5r\frac{d}{dr}\nu(r) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}\sin^2(\theta)\ ,$$

.

2.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^0{}_0 = \frac{\left(0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5r\frac{d^2}{dr^2}\nu(r) - 1.0\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r}\ ,$$

$$R^1{}_1 = \frac{\left(r\left(0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r)\right) + 1.0\frac{d}{dr}\lambda(r)\right)e^{-\lambda(r)}}{r} ,$$

$$R^2{}_2 = \frac{\left(0.5r\left(\frac{d}{dr}\lambda(r) - \frac{d}{dr}\nu(r)\right) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}}{r^2} ,$$

$$R^3{}_3 = \frac{\left(0.5r\frac{d}{dr}\lambda(r) - 0.5r\frac{d}{dr}\nu(r) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}}{r^2} ,$$

.

2.5 Ricci scalar

The Ricci scalar is

$$R = \frac{\left(0.5r^2\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.5r^2\left(\frac{d}{dr}\nu(r)\right)^2 - 1.0r^2\frac{d^2}{dr^2}\nu(r) + 2.0r\frac{d}{dr}\lambda(r) - 2.0r\frac{d}{dr}\nu(r) + 2.0e^{\lambda(r)} - 2.0\right)}{r^2}$$

2.6 $(0,2)$ Einstein tensor

The non-vanishing components of the $(0,2)$ Einstein tensor are

$$G_{00} = \frac{1.0\left(r\frac{d}{dr}\lambda(r) + e^{\lambda(r)} - 1\right)e^{-\lambda(r)+\nu(r)}}{r^2} ,$$

$$G_{11} = \frac{1.0\left(r\frac{d}{dr}\nu(r) - e^{\lambda(r)} + 1\right)}{r^2} ,$$

$$G_{22} = r\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)} ,$$

$$G_{33} = 1.0r\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(}$$

.

2.7 $(1,1)$ Einstein tensor

The non-vanishing components of the $(1,1)$ Einstein tensor are

$$G^0{}_0 = \frac{1.0\left(-r\frac{d}{dr}\lambda(r) - e^{\lambda(r)} + 1\right)e^{-\lambda(r)}}{r^2} ,$$

$$G^1{}_1 = \frac{1.0\left(r\frac{d}{dr}\nu(r) - e^{\lambda(r)} + 1\right)e^{-\lambda(r)}}{r^2} ,$$

$$G^2{}_2 = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r},$$

$$G^3{}_3 = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r},$$

.

Chapter 3

Interior Schwarschild metric (2)

The metric is

$$g = \begin{bmatrix} \frac{(2Gm(r)-r)e^{h(r)}}{r} & 0 & 0 & 0 \\ 0 & \frac{r}{-2Gm(r)+r} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -\frac{r}{-2Gm(r)e^{h(r)}+re^{h(r)}} & 0 & 0 & 0 \\ 0 & -\frac{2Gm(r)}{r}+1 & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

3.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = \frac{-1.0Gm(r) + r\left(1.0G\frac{d}{dr}m(r) + 0.5\cdot(2Gm(r)-r)\frac{d}{dr}h(r) - 0.5\right) + 0.5r}{r\,(2Gm(r)-r)} , \quad \Gamma^0_{10} = \frac{-1.0Gm(r) +}{}$$

$$\Gamma^1_{11} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r\,(-2.0Gm(r)+1.0r)} , \quad \Gamma^1_{22} = 2.0Gm(r) - 1.0r , \quad \Gamma^1_{33} = (2.0Gm(r) - 1.0r)\sin^2(\theta) ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} , \quad \Gamma^2_{21} = \frac{1.0}{r} , \quad \Gamma^2_{33} = -0.5\sin(2\theta) ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} , \quad \Gamma^3_{23} = \frac{1.0}{\tan(\theta)} , \quad \Gamma^3_{31} = \frac{1.0}{r} ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} , \quad .$$

3.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = \frac{0.5Gr^2m(r)\left(\frac{d}{dr}h(r)\right)^2 + 1.0Gr^2m(r)\frac{d^2}{dr^2}h(r) + 1.5Gr^2\frac{d}{dr}h(r)\frac{d}{dr}m(r) + 1.0Gr^2\frac{d^2}{dr^2}m(r) - 1.5Gr}{r^2\left(-2.0Gm(r) + 1.0\right.}$$

$$R^0{}_{110} = \frac{-0.5Gr^2m(r)\left(\frac{d}{dr}h(r)\right)^2 - 1.0Gr^2m(r)\frac{d^2}{dr^2}h(r) - 1.5Gr^2\frac{d}{dr}h(r)\frac{d}{dr}m(r) - 1.0Gr^2\frac{d^2}{dr^2}m(r) + 1.5G}{r^2\left(-2.0Gm(r) + 1.\right.}$$

$$R^0{}_{202} = 1.0Gm(r)\frac{d}{dr}h(r) + 1.0G\frac{d}{dr}m(r) - \frac{1.0Gm(r)}{r} - 0.5r\frac{d}{dr}h(r) \ ,$$

$$R^0{}_{220} = -1.0Gm(r)\frac{d}{dr}h(r) - 1.0G\frac{d}{dr}m(r) + \frac{1.0Gm(r)}{r} + 0.5r\frac{d}{dr}h(r) \ ,$$

$$R^0{}_{303} = \frac{\left(1.0Grm(r)\frac{d}{dr}h(r) + 1.0Gr\frac{d}{dr}m(r) - 1.0Gm(r) - 0.5r^2\frac{d}{dr}h(r)\right)\sin^2(\theta)}{r} \ ,$$

$$R^0{}_{330} = \frac{\left(-1.0Grm(r)\frac{d}{dr}h(r) - 1.0Gr\frac{d}{dr}m(r) + 1.0Gm(r) + 0.5r^2\frac{d}{dr}h(r)\right)\sin^2(\theta)}{r} \ ,$$

$$R^1{}_{001} = \frac{\left(-1.0G^2r^2m^2(r)\left(\frac{d}{dr}h(r)\right)^2 - 2.0G^2r^2m^2(r)\frac{d^2}{dr^2}h(r) - 3.0G^2r^2m(r)\frac{d}{dr}h(r)\frac{d}{dr}m(r) - 2.0G^2r^2m\right.}{}$$

$$R^1{}_{010} = \frac{\left(1.0G^2r^2m^2(r)\left(\frac{d}{dr}h(r)\right)^2 + 2.0G^2r^2m^2(r)\frac{d^2}{dr^2}h(r) + 3.0G^2r^2m(r)\frac{d}{dr}h(r)\frac{d}{dr}m(r) + 2.0G^2r^2m(r\right.}{}$$

$$R^1{}_{212} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r} \ ,$$

$$R^1{}_{221} = \frac{1.0G\left(-r\frac{d}{dr}m(r) + m(r)\right)}{r} \ ,$$

$$R^1{}_{313} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)\sin^2(\theta)}{r} \ ,$$

$$R^1{}_{331} = \frac{1.0G\left(-r\frac{d}{dr}m(r) + m(r)\right)\sin^2(\theta)}{r} \ ,$$

$$R^2{}_{002} = -\frac{1.0\cdot(1.0Gm(r) - 0.5r)\left(-2Gm(r) + r\left(2G\frac{d}{dr}m(r) + (2Gm(r) - r)\frac{d}{dr}h(r) - 1\right) + r\right)e^{h(r)}}{r^4} \ ,$$

$$R^2{}_{020} = \frac{1.0\cdot(1.0Gm(r) - 0.5r)\left(-2Gm(r) + r\left(2G\frac{d}{dr}m(r) + (2Gm(r) - r)\frac{d}{dr}h(r) - 1\right) + r\right)e^{h(r)}}{r^4} \ ,$$

$$R^2{}_{112} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r^2\cdot(2.0Gm(r) - 1.0r)} \ ,$$

$$R^2{}_{121} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r^2\left(-2.0Gm(r) + 1.0r\right)} \ ,$$

$$R^2{}_{323} = \frac{2.0Gm(r)\sin^2(\theta)}{r} \ ,$$

$$R^2{}_{332} = -\frac{2.0Gm(r)\sin^2(\theta)}{r} \ ,$$

$$R^3{}_{003} = -\frac{1.0\cdot(1.0Gm(r) - 0.5r)\left(-2Gm(r) + r\left(2G\frac{d}{dr}m(r) + (2Gm(r) - r)\frac{d}{dr}h(r) - 1\right) + r\right)e^{h(r)}}{r^4} \ ,$$

$$R^3{}_{030} = \frac{1.0\cdot(1.0Gm(r) - 0.5r)\left(-2Gm(r) + r\left(2G\frac{d}{dr}m(r) + (2Gm(r) - r)\frac{d}{dr}h(r) - 1\right) + r\right)e^{h(r)}}{r^4} \ ,$$

$$R^3{}_{113} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r^2\cdot(2.0Gm(r) - 1.0r)} \ ,$$

$$R^3{}_{131} = \frac{1.0G\left(r\frac{d}{dr}m(r) - m(r)\right)}{r^2\left(-2.0Gm(r) + 1.0r\right)} \ ,$$

$$R^3{}_{223} = -\frac{2.0Gm(r)}{r} \ ,$$

$$R^3{}_{232} = \frac{2.0Gm(r)}{r} \ ,$$

.

3.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = \frac{\left(1.0G^2rm^2(r)\left(\frac{d}{dr}h(r)\right)^2 + 2.0G^2rm^2(r)\frac{d^2}{dr^2}h(r) + 3.0G^2rm(r)\frac{d}{dr}h(r)\frac{d}{dr}m(r) + 2.0G^2rm(r)\frac{d^2}{dr^2}m\right.}{}$$

$$R_{11} = \frac{0.5Grm(r)\left(\frac{d}{dr}h(r)\right)^2 + 1.0Grm(r)\frac{d^2}{dr^2}h(r) + 1.5Gr\frac{d}{dr}h(r)\frac{d}{dr}m(r) + 1.0Gr\frac{d^2}{dr^2}m(r) - 1.5Gm(r)\frac{d}{d}}{r\left(-2.0Gm(r) + 1.0r\right)}$$

$$R_{22} = 1.0Gm(r)\frac{d}{dr}h(r) + 2.0G\frac{d}{dr}m(r) - 0.5r\frac{d}{dr}h(r) \ ,$$

$$R_{33} = \left(1.0Gm(r)\frac{d}{dr}h(r) + 2.0G\frac{d}{dr}m(r) - 0.5r\frac{d}{dr}h(r)\right)\sin^2(\theta) \ ,$$

.

3.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^0{}_0 = \frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} + \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} + \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} + \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{0.5Gm(r)\frac{d}{dr}h(r)}{r^2} -$$

$$R^1{}_1 = \frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} + \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} + \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} + \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} - \frac{1.5Gm(r)\frac{d}{dr}h(r)}{r^2} -$$

$$R^2{}_2 = \frac{1.0Gm(r)\frac{d}{dr}h(r) + 2.0G\frac{d}{dr}m(r) - 0.5r\frac{d}{dr}h(r)}{r^2} ,$$

$$R^3{}_3 = \frac{1.0Gm(r)\frac{d}{dr}h(r) + 2.0G\frac{d}{dr}m(r) - 0.5r\frac{d}{dr}h(r)}{r^2} ,$$

.

3.5 Ricci scalar

The Ricci scalar is

$$R = \frac{1.0Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} + \frac{2.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} + \frac{3.0G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} + \frac{2.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{1.0Gm(r)\frac{d}{dr}h(r)}{r^2} + 4$$

3.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{1.0G\left(-4.0Gm(r) + 2.0r\right)e^{h(r)}\frac{d}{dr}m(r)}{r^3} ,$$

$$G_{11} = \frac{-2.0Gm(r)\frac{d}{dr}h(r) - 2.0G\frac{d}{dr}m(r) + 1.0r\frac{d}{dr}h(r)}{r\left(-2.0Gm(r) + 1.0r\right)} ,$$

$$G_{22} = -0.5Grm(r)\left(\frac{d}{dr}h(r)\right)^2 - 1.0Grm(r)\frac{d^2}{dr^2}h(r) - 1.5Gr\frac{d}{dr}h(r)\frac{d}{dr}m(r) - 1.0Gr\frac{d^2}{dr^2}m(r) + 0.5Gm(r$$

$$G_{33} = \left(-0.5Grm(r)\left(\frac{d}{dr}h(r)\right)^2 - 1.0Grm(r)\frac{d^2}{dr^2}h(r) - 1.5Gr\frac{d}{dr}h(r)\frac{d}{dr}m(r) - 1.0Gr\frac{d^2}{dr^2}m(r) + 0.5\right.$$

.

3.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^{0}{}_{0} = -\frac{2.0G\frac{d}{dr}m(r)}{r^2} \ ,$$

$$G^{1}{}_{1} = \frac{1.0\left(-2.0Gm(r)\frac{d}{dr}h(r) - 2.0G\frac{d}{dr}m(r) + 1.0r\frac{d}{dr}h(r)\right)}{r^2} \ ,$$

$$G^{2}{}_{2} = -\frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} - \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} - \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} - \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{0.5Gm(r)\frac{d}{dr}h(r)}{r^2}$$

$$G^{3}{}_{3} = -\frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} - \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} - \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} - \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{0.5Gm(r)\frac{d}{dr}h(r)}{r^2}$$

.

Chapter 4

Interior Schwarschild metric (3)

The metric is

$$g = \begin{bmatrix} -f(r)h(r) & 0 & 0 & 0 \\ 0 & \frac{1}{f(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -\frac{1}{f(r)h(r)} & 0 & 0 & 0 \\ 0 & f(r) & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

4.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = \frac{0.5\frac{d}{dr}h(r)}{h(r)} + \frac{0.5\frac{d}{dr}f(r)}{f(r)} , \quad \Gamma^0_{10} = \frac{0.5\frac{d}{dr}h(r)}{h(r)} + \frac{0.5\frac{d}{dr}f(r)}{f(r)} , \quad \Gamma^1_{00} = 0.5\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)$$

$$\Gamma^1_{11} = -\frac{0.5\frac{d}{dr}f(r)}{f(r)} , \quad \Gamma^1_{22} = -1.0rf(r) , \quad \Gamma^1_{33} = -1.0rf(r)\sin^2(\theta) ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} , \quad \Gamma^2_{21} = \frac{1.0}{r} , \quad \Gamma^2_{33} = -0.5\sin(2\theta) ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} , \quad \Gamma^3_{23} = \frac{1.0}{\tan(\theta)} , \quad \Gamma^3_{31} = \frac{1.0}{r} ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} , \quad .$$

4.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = -\frac{0.5\frac{d^2}{dr^2}h(r)}{h(r)} + \frac{0.25\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)} - \frac{0.5\frac{d^2}{dr^2}f(r)}{f(r)} - \frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{f(r)h(r)} \ ,$$

$$R^0{}_{110} = \frac{0.5\frac{d^2}{dr^2}h(r)}{h(r)} - \frac{0.25\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)} + \frac{0.5\frac{d^2}{dr^2}f(r)}{f(r)} + \frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{f(r)h(r)} \ ,$$

$$R^0{}_{202} = -\frac{0.5r\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)}{h(r)} \ ,$$

$$R^0{}_{220} = \frac{0.5r\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)}{h(r)} \ ,$$

$$R^0{}_{303} = -\frac{0.5r\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)\sin^2(\theta)}{h(r)} \ ,$$

$$R^0{}_{330} = \frac{0.5r\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)\sin^2(\theta)}{h(r)} \ ,$$

$$R^1{}_{001} = -0.5f^2(r)\frac{d^2}{dr^2}h(r) + \frac{0.25f^2(r)\left(\frac{d}{dr}h(r)\right)^2}{h(r)} - 0.5f(r)h(r)\frac{d^2}{dr^2}f(r) - 0.75f(r)\frac{d}{dr}f(r)\frac{d}{dr}h(r) \ ,$$

$$R^1{}_{010} = 0.5f^2(r)\frac{d^2}{dr^2}h(r) - \frac{0.25f^2(r)\left(\frac{d}{dr}h(r)\right)^2}{h(r)} + 0.5f(r)h(r)\frac{d^2}{dr^2}f(r) + 0.75f(r)\frac{d}{dr}f(r)\frac{d}{dr}h(r) \ ,$$

$$R^1{}_{212} = -0.5r\frac{d}{dr}f(r) \ ,$$

$$R^1{}_{221} = 0.5r\frac{d}{dr}f(r) \ ,$$

$$R^1{}_{313} = -0.5r\sin^2(\theta)\frac{d}{dr}f(r) \ ,$$

$$R^1{}_{331} = 0.5r\sin^2(\theta)\frac{d}{dr}f(r) \ ,$$

$$R^2{}_{002} = -\frac{0.5\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)f(r)}{r} \ ,$$

$$R^2{}_{020} = \frac{0.5\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)f(r)}{r} \ ,$$

$$R^2{}_{112} = \frac{0.5\frac{d}{dr}f(r)}{rf(r)} \ ,$$

$$R^2{}_{121} = -\frac{0.5\frac{d}{dr}f(r)}{rf(r)} \ ,$$

$$R^2{}_{323} = (1.0 - 1.0f(r))\sin^2(\theta) \ ,$$

$$R^2{}_{332} = (1.0f(r) - 1.0)\sin^2(\theta) \ ,$$

$$R^3{}_{003} = -\frac{0.5\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)f(r)}{r} \ ,$$

$$R^3{}_{030} = \frac{0.5\left(f(r)\frac{d}{dr}h(r) + h(r)\frac{d}{dr}f(r)\right)f(r)}{r} \ ,$$

$$R^3{}_{113} = \frac{0.5\frac{d}{dr}f(r)}{rf(r)} \ ,$$

$$R^3{}_{131} = -\frac{0.5\frac{d}{dr}f(r)}{rf(r)} \ ,$$

$$R^3{}_{223} = 1.0f(r) - 1.0 \ ,$$

$$R^3{}_{232} = 1.0 - 1.0f(r) \ ,$$

.

4.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = \frac{\left(r\left(0.5f(r)\frac{d^2}{dr^2}h(r) + 0.5h(r)\frac{d^2}{dr^2}f(r) + 0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)\right)h(r) - 0.25rf(r)\left(\frac{d}{dr}h(r)\right)^2 + 1.0\left(f(\right.\right.}{rh(r)}$$

$$R_{11} = -\frac{0.5\frac{d^2}{dr^2}h(r)}{h(r)} + \frac{0.25\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)} - \frac{0.5\frac{d^2}{dr^2}f(r)}{f(r)} - \frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{f(r)h(r)} - \frac{1.0\frac{d}{dr}f(r)}{rf(r)} \ ,$$

$$R_{22} = -\frac{0.5rf(r)\frac{d}{dr}h(r)}{h(r)} - 1.0r\frac{d}{dr}f(r) - 1.0f(r) + 1.0 \ ,$$

$$R_{33} = \frac{\left(-0.5rf(r)\frac{d}{dr}h(r) - 1.0rh(r)\frac{d}{dr}f(r) - 1.0f(r)h(r) + 1.0h(r)\right)\sin^2(\theta)}{h(r)} \ ,$$

.

4.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^{0}{}_{0} = -\frac{0.5f(r)\frac{d^2}{dr^2}h(r)}{h(r)}+\frac{0.25f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)}-0.5\frac{d^2}{dr^2}f(r)-\frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)}-\frac{1.0f(r)\frac{d}{dr}h(r)}{rh(r)}-\frac{1.0\frac{d}{dr}f(}{r}$$

$$R^{1}{}_{1} = -\frac{0.5f(r)\frac{d^2}{dr^2}h(r)}{h(r)}+\frac{0.25f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)}-0.5\frac{d^2}{dr^2}f(r)-\frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)}-\frac{1.0\frac{d}{dr}f(r)}{r} ,$$

$$R^{2}{}_{2} = \frac{-0.5rf(r)\frac{d}{dr}h(r) - 1.0\left(r\frac{d}{dr}f(r) + f(r) - 1\right)h(r)}{r^2h(r)} ,$$

$$R^{3}{}_{3} = -\frac{0.5f(r)\frac{d}{dr}h(r)}{rh(r)} - \frac{1.0\frac{d}{dr}f(r)}{r} - \frac{1.0f(r)}{r^2} + \frac{1.0}{r^2} ,$$

.

4.5 Ricci scalar

The Ricci scalar is

$$R = -\frac{1.0f(r)\frac{d^2}{dr^2}h(r)}{h(r)}+\frac{0.5f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)}-1.0\frac{d^2}{dr^2}f(r)-\frac{1.5\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)}-\frac{2.0f(r)\frac{d}{dr}h(r)}{rh(r)}-\frac{4.0\frac{d}{dr}f(r)}{r}-$$

4.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{1.0\left(-r\frac{d}{dr}f(r) - f(r) + 1\right)f(r)h(r)}{r^2} ,$$

$$G_{11} = \frac{1.0\frac{d}{dr}h(r)}{rh(r)} + \frac{1.0\frac{d}{dr}f(r)}{rf(r)} + \frac{1.0}{r^2} - \frac{1.0}{r^2f(r)} ,$$

$$G_{22} = \frac{0.5r^2f(r)\frac{d^2}{dr^2}h(r)}{h(r)}-\frac{0.25r^2f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)}+0.5r^2\frac{d^2}{dr^2}f(r)+\frac{0.75r^2\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)}+\frac{0.5rf(r)\frac{d}{dr}h(r)}{h(r)}+$$

$$G_{33} = \frac{1.0r\left(0.5rf(r)h(r)\frac{d^2}{dr^2}h(r) - 0.25rf(r)\left(\frac{d}{dr}h(r)\right)^2 + 0.5rh^2(r)\frac{d^2}{dr^2}f(r) + 0.75rh(r)\frac{d}{dr}f(r)\frac{d}{dr}h(r) +\right.}{h^2(r)}$$

.

4.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^{0}{}_{0} = \frac{1.0\left(r\frac{d}{dr}f(r) + f(r) - 1\right)}{r^2} \ ,$$

$$G^{1}{}_{1} = \frac{1.0f(r)\frac{d}{dr}h(r)}{rh(r)} + \frac{1.0\frac{d}{dr}f(r)}{r} + \frac{1.0f(r)}{r^2} - \frac{1.0}{r^2} \ ,$$

$$G^{2}{}_{2} = \frac{0.5f(r)\frac{d^2}{dr^2}h(r)}{h(r)} - \frac{0.25f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)} + 0.5\frac{d^2}{dr^2}f(r) + \frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)} + \frac{0.5f(r)\frac{d}{dr}h(r)}{rh(r)} + \frac{1.0\frac{d}{dr}f(r}{r}$$

$$G^{3}{}_{3} = \frac{0.5f(r)\frac{d^2}{dr^2}h(r)}{h(r)} - \frac{0.25f(r)\left(\frac{d}{dr}h(r)\right)^2}{h^2(r)} + 0.5\frac{d^2}{dr^2}f(r) + \frac{0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r)}{h(r)} + \frac{0.5f(r)\frac{d}{dr}h(r)}{rh(r)} + \frac{1.0\frac{d}{dr}f(r}{r}$$

.

Chapter 5

FRW metric

The metric is

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -\frac{a^2(t)}{kr^2-1} & 0 & 0 \\ 0 & 0 & r^2a^2(t) & 0 \\ 0 & 0 & 0 & r^2a^2(t)\sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{-kr^2+1}{a^2(t)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2a^2(t)} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2a^2(t)\sin^2(\theta)} \end{bmatrix} ,$$

.

5.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{11} = -\frac{1.0a(t)\frac{d}{dt}a(t)}{kr^2-1} , \quad \Gamma^0_{22} = 1.0r^2a(t)\frac{d}{dt}a(t) , \quad \Gamma^0_{33} = 1.0r^2a(t)\sin^2(\theta)\frac{d}{dt}a(t) ,$$

$$\Gamma^1_{01} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^1_{10} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^1_{11} = -\frac{1.0kr}{kr^2-1} ,$$

$$\Gamma^1_{22} = 1.0r\left(kr^2-1\right) , \quad \Gamma^1_{33} = 1.0r\left(kr^2-1\right)\sin^2(\theta) , \quad \Gamma^2_{02} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} , \quad \Gamma^2_{20} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^2_{21} = \frac{1.0}{r} ,$$

$$\Gamma^2_{33} = -0.5\sin(2\theta)\ , \quad \Gamma^3_{03} = \frac{1.0\frac{d}{dt}a(t)}{a(t)}\ , \quad \Gamma^3_{13} = \frac{1.0}{r}\ ,$$

$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)}\ , \quad \Gamma^3_{30} = \frac{1.0\frac{d}{dt}a(t)}{a(t)}\ , \quad \Gamma^3_{31} = \frac{1.0}{r}\ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)}\ , \quad .$$

5.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = -\frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2-1}\ ,$$

$$R^0{}_{110} = \frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2-1}\ ,$$

$$R^0{}_{202} = 1.0r^2a(t)\frac{d^2}{dt^2}a(t)\ ,$$

$$R^0{}_{220} = -1.0r^2a(t)\frac{d^2}{dt^2}a(t)\ ,$$

$$R^0{}_{303} = 1.0r^2a(t)\sin^2(\theta)\frac{d^2}{dt^2}a(t)\ ,$$

$$R^0{}_{330} = -1.0r^2a(t)\sin^2(\theta)\frac{d^2}{dt^2}a(t)\ ,$$

$$R^1{}_{001} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)}\ ,$$

$$R^1{}_{010} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)}\ ,$$

$$R^1{}_{212} = 1.0r^2\left(k+\left(\frac{d}{dt}a(t)\right)^2\right)\ ,$$

$$R^1{}_{221} = 1.0r^2\left(-k-\left(\frac{d}{dt}a(t)\right)^2\right)\ ,$$

$$R^1{}_{313} = 1.0r^2\left(k+\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta)\ ,$$

$$R^1{}_{331} = 1.0r^2\left(-k-\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta)\ ,$$

$$R^2{}_{002} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^2{}_{020} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^2{}_{112} = \frac{1.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^2{}_{121} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^2{}_{323} = 1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^2{}_{332} = -1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^3{}_{003} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^3{}_{030} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^3{}_{113} = \frac{1.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^3{}_{131} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^3{}_{223} = 1.0r^2\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R^3{}_{232} = 1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

.

5.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = -\frac{3.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R_{11} = \frac{-2.0k - 1.0a(t)\frac{d^2}{dt^2}a(t) - 2.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$

$$R_{22} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R_{33} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

.

5.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^0{}_0 = \frac{3.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^1{}_1 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$R^2{}_2 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$R^3{}_3 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

.

5.5 Ricci scalar

The Ricci scalar is

$$R = \frac{6.0\left(k + a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ .$$

5.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{3.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ ,$$

$$G_{11} = \frac{1.0k + 2.0a(t)\frac{d^2}{dt^2}a(t) + 1.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$

$$G_{22} = r^2\left(-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$G_{33} = r^2\left(-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

.

5.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^0{}_0 = \frac{3.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ ,$$

$$G^1{}_1 = \frac{-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$G^2{}_2 = \frac{-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$G^3{}_3 = \frac{-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

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Bibliography

[1] https://github.com/PhysicsZandi/GeneralRelativity/blob/main/source/py/python_gr.tex.